INTRODUCTION TO INFORMATION GEOMETRY

Probability Laws, Means, and Variances

Week 4 · Lecture 1

Christopher Lee

A probability law is a mathematical state

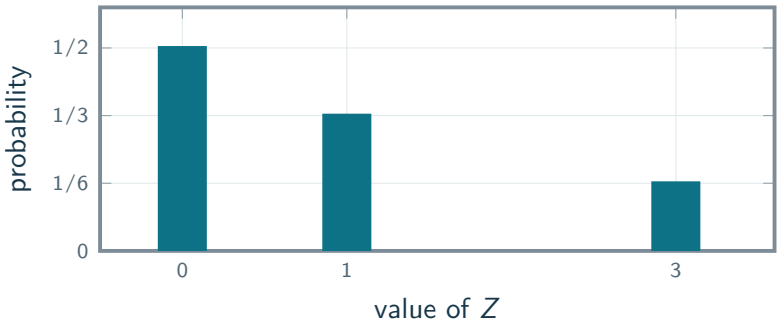
For three possible outcomes,

$$P = (p_A, p_B, p_C), \qquad p_i \geq 0, \quad \sum_i p_i = 1.$$

The vector $(0.5, 0.3, 0.2)$ describes the probabilities of all possible results of one observation. A single sample instead produces one outcome, such as B.

Probabilities and numerical outcomes

Let Z take values $0, 1, 3$ with probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$.



Expected value as a function of probabilities

For fixed numerical values z_i, define

$$E_P[Z] = \sum_i z_i p_i.$$

Changing P changes the expectation even if the possible values of Z remain the same.

This makes expectation a real-valued function on the simplex.

The mean of a three-outcome law

For $z = (0, 1, 3)$ and $P = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$,

$$E_P[Z] = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{3} + 3 \cdot \frac{1}{6}.$$

Find the expected value: $\mu = ?$.

The mean of a three-outcome law

For $z = (0, 1, 3)$ and $P = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$,

$$E_P[Z] = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{3} + 3 \cdot \frac{1}{6}.$$

The expected value is $\mu = \frac{5}{6}$.

Variance measures spread

Write $\mu = E_P[Z]$. The variance is

$$\mathrm{Var}_P(Z) = \sum_i p_i (z_i - \mu)^2.$$

Equivalently,

$$\mathrm{Var}_P(Z) = E_P[Z^2] - \mu^2.$$

It is another function on the space of distributions.

The variance of the example

For $z = (0, 1, 3)$, $P = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$, and $\mu = \frac{5}{6}$, compute

$$E[Z^2] = 0 + \frac{1}{3} + \frac{9}{6} = ?.$$

$$\mathrm{Var}(Z) = E[Z^2] - \mu^2 = ?.$$

The variance of the example

For $z = (0, 1, 3)$, $P = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$, and $\mu = \frac{5}{6}$,

$$E[Z^2] = 0 + \frac{1}{3} + \frac{9}{6} = \frac{11}{6}.$$

$$\mathrm{Var}(Z) = \frac{11}{6} - \frac{25}{36} = \frac{41}{36}.$$

The Bernoulli family

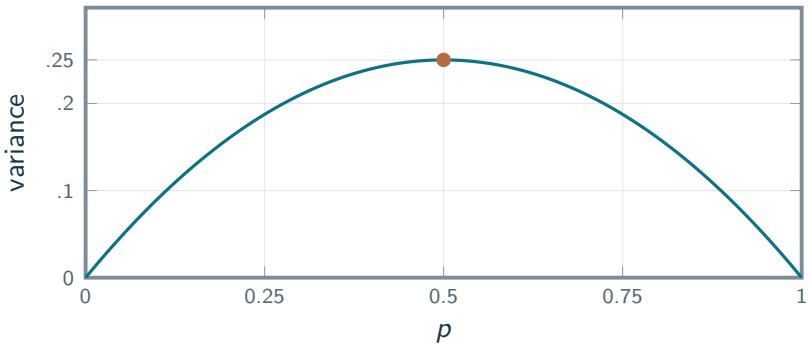
A Bernoulli random variable takes values 0 and 1.

$$P_p = (1-p, p), \qquad 0 \leq p \leq 1.$$

The parameter p determines the whole distribution:

$$E[X] = p, \qquad \mathrm{Var}(X) = p(1-p).$$

Variance along the Bernoulli interval



The variance is largest at $p = \frac{1}{2}$ and vanishes at the pure states.

Expectation on the probability triangle

Set $P = (x, y, 1 - x - y)$ and $z = (0, 1, 3)$.

$$\mu(x, y) = 0x + y + 3(1 - x - y) = 3 - 3x - 2y.$$

$$\frac{\partial \mu}{\partial x} = -3, \qquad \frac{\partial \mu}{\partial y} = -2.$$

Probability laws supply an ordinary domain for multivariable calculus.

Moving weight among outcomes

Let $z = (0, 1, 3)$ and $P = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$. Compute the expectation:

$$E_P[Z] = ?.$$

Compare with the earlier distribution $(\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$.

Moving weight among outcomes

Let $z = (0, 1, 3)$ and $P = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$. The expectation is

$$E_P[Z] = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 3 \cdot \frac{1}{4} = \frac{5}{4}.$$

It exceeds the earlier mean $\frac{5}{6}$: more probability lies on larger values.

A state and a sample answer different questions

A probability distribution specifies the model's possible outcomes and their chances. An observation records what occurred on one trial. An empirical distribution records the proportions of outcomes in a collection of trials. These objects can all be drawn in the same simplex.